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Started	Friday, 30 October 2026, 10:58 PM
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Duration	6 mins 33 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Complete

Mark 1.00 out of 1.00

What is Bubble Sort known for?

a. Using the divide-and-conquer approach
b. Sorting data in a non-sequential manner
c. Being the most efficient sorting algorithm
d. Bubbling up the largest element to its correct position with each pass

The correct answer is: Bubbling up the largest element to its correct position with each pass

Question **2**

Complete

Mark 0.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

a. Bubble Sort
b. Merge Sort
c. Quick Sort
d. Linear Search

The correct answer is: Merge Sort

Question **3**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

a. It makes the data unsorted
b. It makes it easier to select items based on their relationship to the rest of the items
c. It complicates the selection of items
d. It slows down the process

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **4**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

a. arrange()
b. order()
c. sort()
d. sorted()

The correct answer is: sort()

Question **5**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

a. Sorting data in a single pass
b. Dividing the input into parts, solving each part, and combining the solutions
c. Sorting data sequentially
d. Rearranging data without sorting

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions

Question **6**

Complete

Mark 1.00 out of 1.00

_____is putting an element in the appropriate place in a sorted list yields a larger sorted order list.

- a. Insertion
- b. Distribution
- c. Extraction
- d. Selection

The correct answer is:
Insertion

Question **7**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

- a. Bubble Sort
- b. Linear Search
- c. Merge Sort
- d. Binary Search

The correct answer is: Merge Sort

Question **8**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

- a. By comparing elements sequentially
- b. By using the bubble-up method
- c. By sorting data in a single pass
- d. By breaking the input into smaller parts and merging them

The correct answer is: By breaking the input into smaller parts and merging them

Question **9**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

- a. Provides a basis for other algorithms to work efficiently
- b. Decreases the efficiency of algorithms
- c. Makes code execution slower
- d. Makes data harder to manage

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **10**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

- a. 200,470,80,150,40,90,300,400,70,120
- b. 80,150,200,470,40,90,300,400,70,120
- c. 40,80,90,150,200,300,400,470,70,120
- d. 40,70,80,90,120,150,200,300,400,470

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **11**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

- a. Combining sorted sublists
- b. Comparing each element with the others
- c. Sorting the entire list sequentially
- d. Dividing the input into smaller subproblems

The correct answer is: Dividing the input into smaller subproblems

Question **12**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

a. Quick Sort
b. Merge Sort
c. Insertion Sort
d. Bubble Sort

The correct answer is: Merge Sort

Question **13**

Complete

Mark 1.00 out of 1.00

What is mean by stable sorting algorithm?

a. A sorting algorithm is stable if it preserves the order of all keys
b. A sorting algorithm is stable if it preserves the order of non-duplicate keys
c. A sorting algorithm is stable if it doesn't preserver the order of duplicate keys
d. A sorting algorithm is stable if it preserves the order of duplicate keys

The correct answer is:
A sorting algorithm is stable if it preserves the order of duplicate keys

Question **14**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

a. Bubbling up the smallest element
b. Sorting the entire list in one pass
c. Bubbling up the largest element to its correct position
d. Dividing the list into halves

The correct answer is: Bubbling up the largest element to its correct position

Question **15**

Complete

Mark 1.00 out of 1.00

The process of placing or rearranging a collection of elements into a particular order is known as

a. Rearranging

b. Merging

c. Searching

d. Sorting

The correct answer is: Sorting